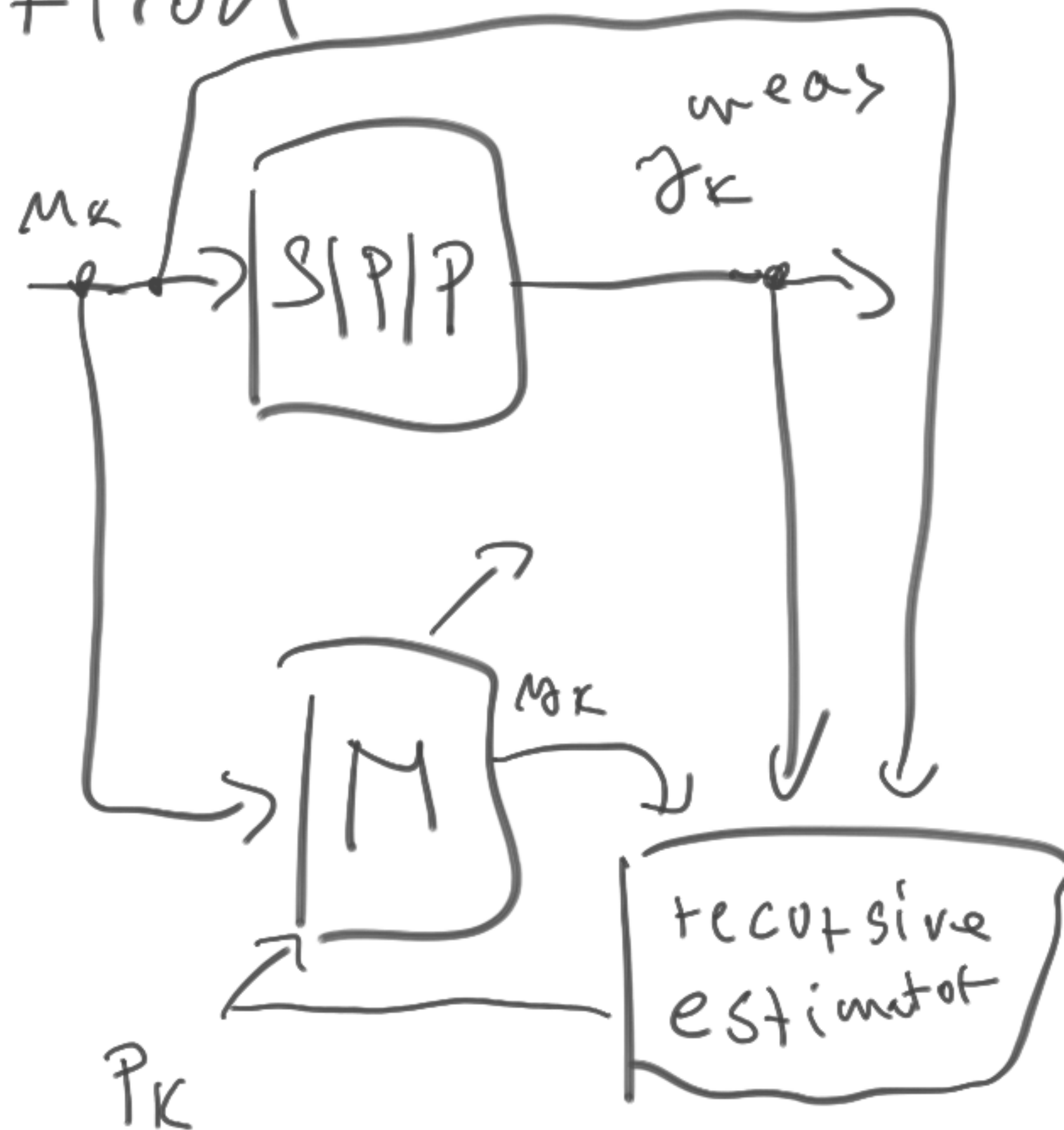
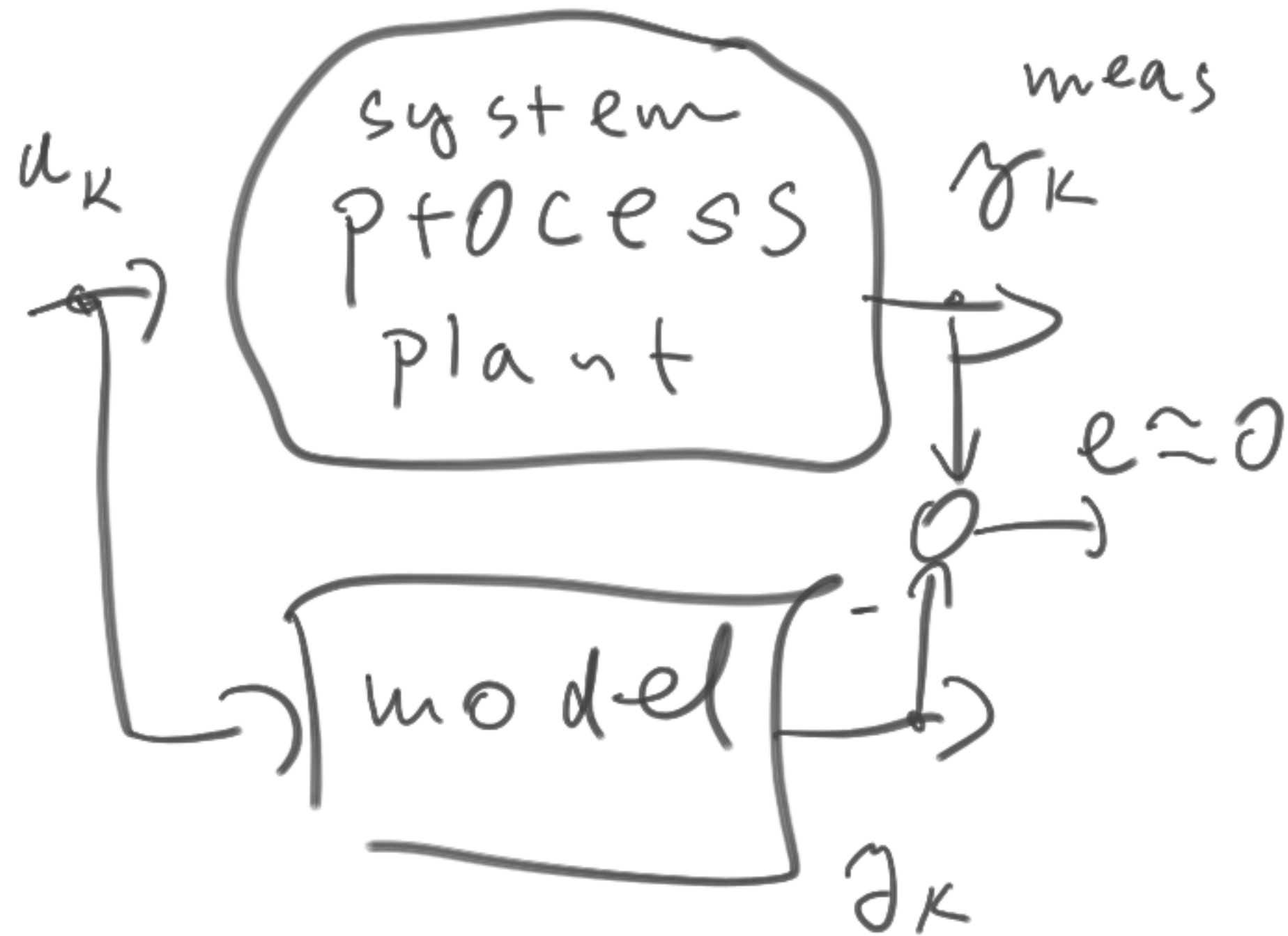


Recursive Estimation



$$y_k = a x_k$$



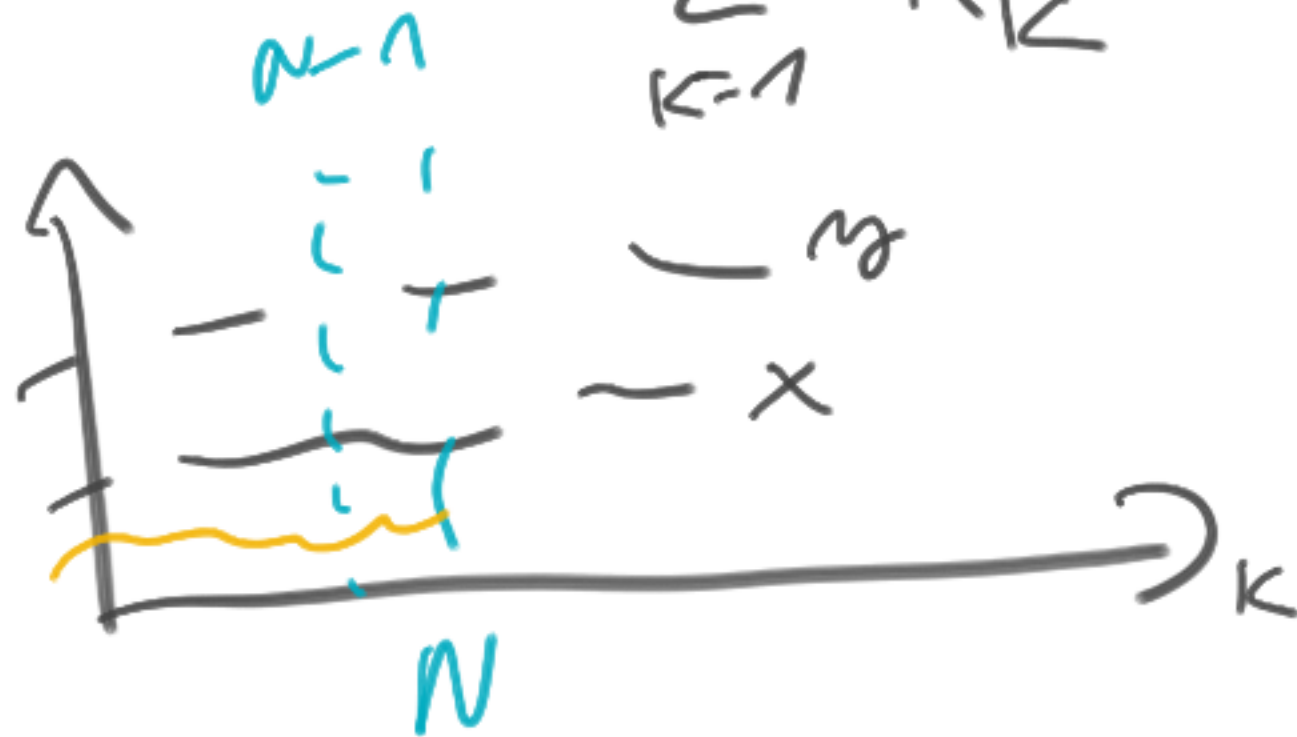
$$a_N =$$

$$\frac{\sum_{k=1}^N y_k x_k}{\sum_{k=1}^N x_k^2}$$

$$a_N = \frac{\sum_{k=1}^{N-1} y_k x_k + y_N x_N}{\sum_{k=1}^N x_k^2}$$

$$a_N = \frac{a_{N-1} \sum_{k=1}^{N-1} x_k^2 + y_N x_N \pm a_{N-1} x_N^2}{\sum_{k=1}^N x_k^2}$$

$$a_{N-1} = \frac{\sum_{k=1}^{N-1} y_k x_k}{\sum_{k=1}^{N-1} x_k^2}$$



$$a_N = \frac{a_{N-1} \sum_{k=1}^N x_k^2 + y_N x_N - a_{N-1} x_N^2}{\sum_{k=1}^N x_k^2}$$

based on information available at time $N-1$

$$\sum_{k=1}^N x_k^2$$

estimation error

prediction

$y_{N|N-1}$

$$a_N = a_{N-1} + \frac{x_N}{\sum_{k=1}^N x_k^2} (y_N - a_{N-1} x_N)$$

$$u_k = K e_k = K (w_k - y_k) \quad \text{controller}$$

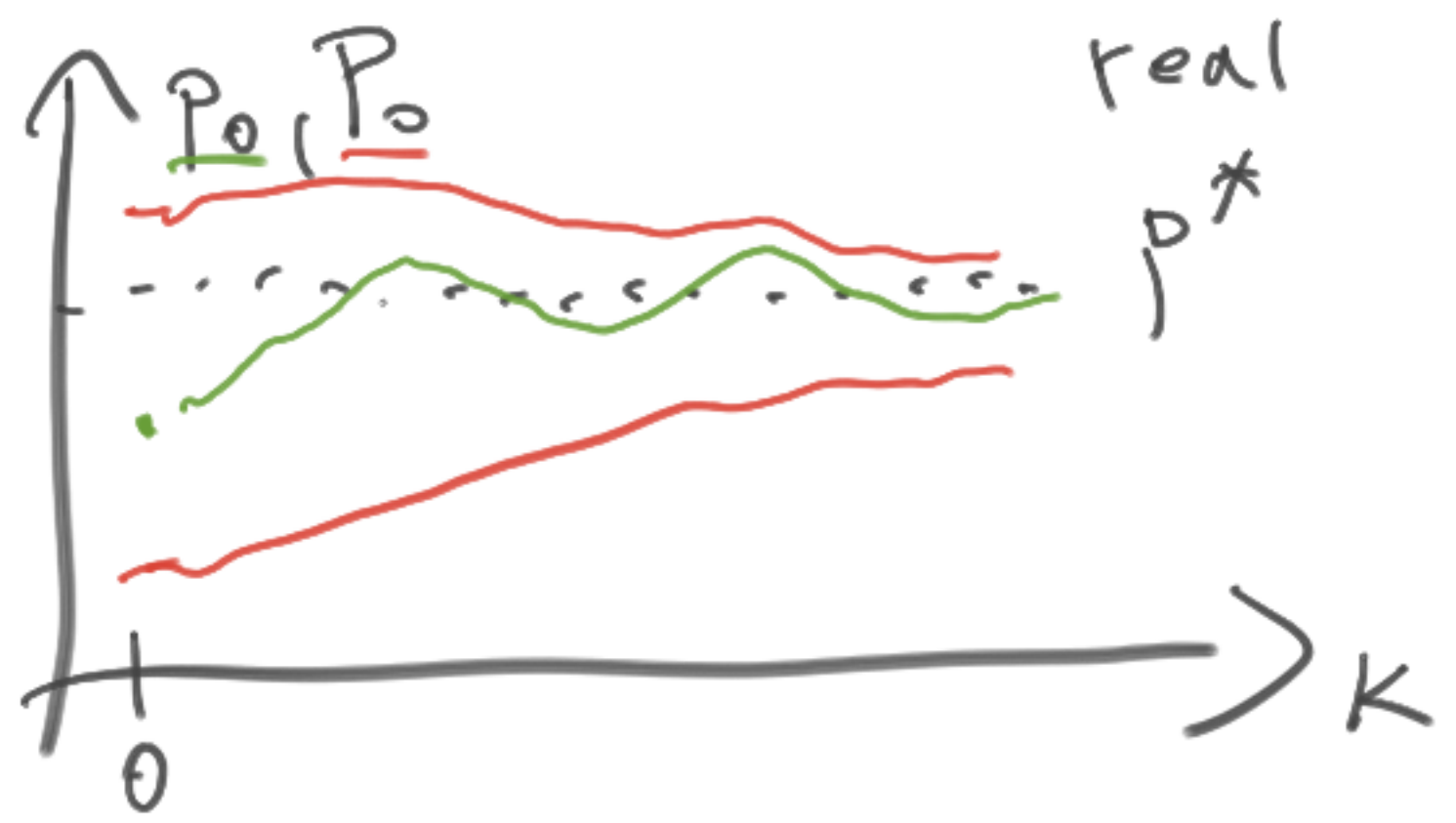
control error

$$y = X P$$

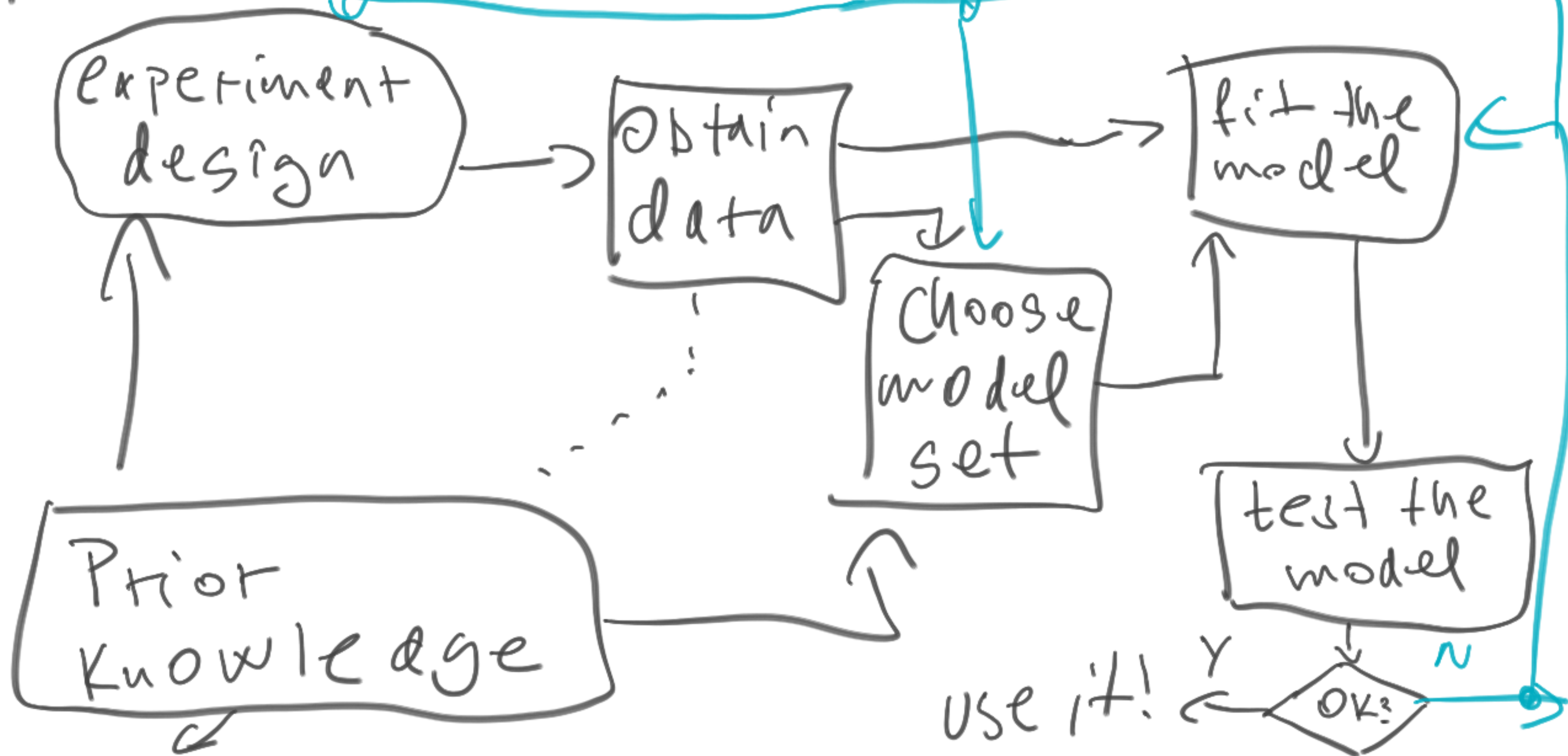
$$y_N = \phi_N^T P$$

$$P_N = P_{N-1} + P_{N-1} \phi_N \left[y_N - \phi_N^T P_{N-1} \right]$$

$$P_N = P_{N-1} + \phi_N \phi_N^T \rightarrow \text{covariance}$$



Practical aspect of ID



How to choose order of a dynamic model?

→ autocorrelation function (ACF)

$$ACF(i) = \frac{1}{N} \sum_{k=1}^N x_k x_{k-i}$$

→ partial autocorrelation fcn. (PACF)

→ cross-correlation (CCF)

$$x_k \mu_k$$